



Optimized Large Components Replacement Process of LMT/OHT

**Maintenance and
Repair**

Application

**Component Life
Management**

**Component
Renewal (CRC)**

**MARC
Management**

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1.0 Introduction

The timing of the PCR of the powertrain components is one of the main factors that impact the availability and performance of the equipment, due to the high downtime required for execution. Smaller components can also significantly impact performance indicators and should be taken into account to meet site maintenance strategies.

Therefore, all possibilities for mitigating the impact on the replacement process of these components must be considered.

2.0 Best Practice Description

This BP describes the strategy adopted by this customer to optimize the components' replacement time through the pre-assembly of component peripherals in a prepared maintenance box with dedicated people to carry out the activity. So, the new components will be ready for installation as soon as the previous one is removed, using the pit stop methodology to perform these activities.

3.0 Implementation Steps

3.1 Area and layout definition of the component box:

The defined area must be strategically located close to the PCR execution box of the components in order to facilitate the load handling logistics and optimization of the performers' travel time. It is necessary that the layout allows for the best arrangement of components for pre-assembly of peripherals. This can be standardized with the preparation and demarcation of the layout of the work area in a clean, safe environment, with enough favorable space for the movement of people and equipment and without interference between maintenance teams, as well as a good contamination control.



Figure 1, 2. Components Pre-assembly Boxes

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3.2 Manufacture of specific supports (Stands) for each component:

In order to carry out the activities safely in a timely manner, suitable devices (brackets, benches, stands etc) must be built in order to support the peripherals for the assembly of these components in a safe and reliable way. They must be made according to the needs of each component and equipment.



Figure 3, 4. Engine stands

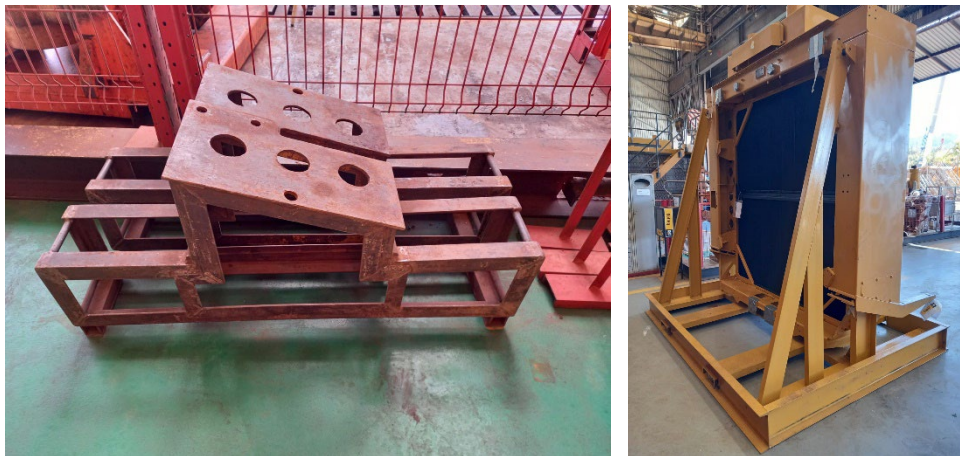


Figure 5,6. Radiator stands



Figure 7, 8. Bench and Support for Front Suspension Assembly and Lift

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3.3 Acquisition of resources and tools:

A list of all the basic and special tools necessary for the assembly and installation of the peripherals for each major component is carried out. Lift devices and hand tools listed must meet the specific demand for preparing these components, without including unnecessary tools in the process.



Figure 9. Hand Tools Board Figure



Figure 10. Special Tools

3.4 Listing/purchasing peripheral kits to install on the engine and radiator:

A list of all the parts and materials necessary for the installation of the components to be prepared must be made. These parts must be available and prepared for installation in the components, reducing downtime for maintenance technicians in the search for these parts and necessary resources.



Figure 11, 12. Parts needed for preassembly process

3.5 Alignment between execution and planning areas:

All resources, parts, labor must be available according to prior planning in order to optimize the execution time similar to a pit stop when replacing components during scheduled maintenance of equipment.

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4.0 Benefits

4.1 Services quality improvement (provide efficient utilization of man hour and resource):

With all the necessary resources available, it is possible to perform a quality service applying all the recommendations specified by the manufacturer, such as adequate torques, layouts and proper routing of hydraulic hoses and electrical harnesses, correct positioning of clamps, among others.

4.2 Contamination control improvement:

Working in a planned and organized way, there is enough time to carry out an effective cleaning, good preparation of the parts, as well as the use of appropriate chemical compounds during the assembly of these in the engine.

4.3 Safety:

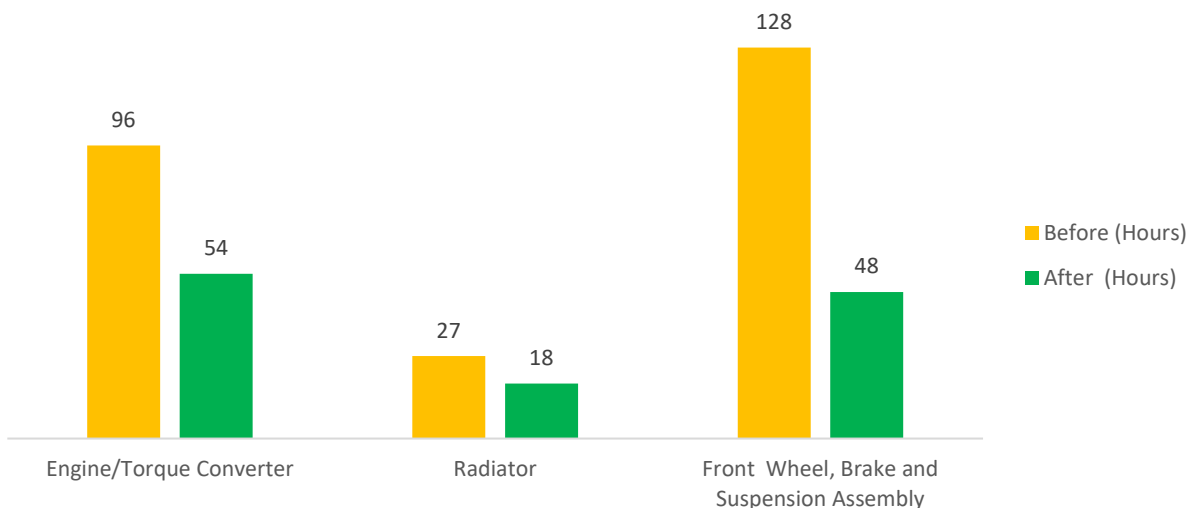
Eliminates the risks of injuries, working in an appropriate place, on a flat floor, without obstructions and properly demarcated, eliminating the technician's exposure to the inherent risks of other activities.

4.4 Time reduction of major R&I components:

This methodology significantly optimizes the time required between the removal and subsequent installation of larger components such as the engine/converter, radiator, suspension and front wheel assembly, with a consequent reduction in machine downtime.

Optimized R&I Components	Before (Hours)	After (Hours)	Gain (Hours)
Engine/Torque Converter	96	54	42
Radiator	27	18	9
Front Wheel, Brake and Suspension Assembly	128	48	80

Table – 01: Time reduction of major R&I components



Graphic – 01: Time reduction of major R&I components

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4.4.1 Engine/Torque Converter:

Engine replacement time on an LMT is reduced by 42 hours (table / graphic 01) as the components are pre-assembled by performing:

- Torque converter installation;
- Installation of the front bar with its subcomponents;
- Installation of front and rear mounts;
- Installation of cooling lines;

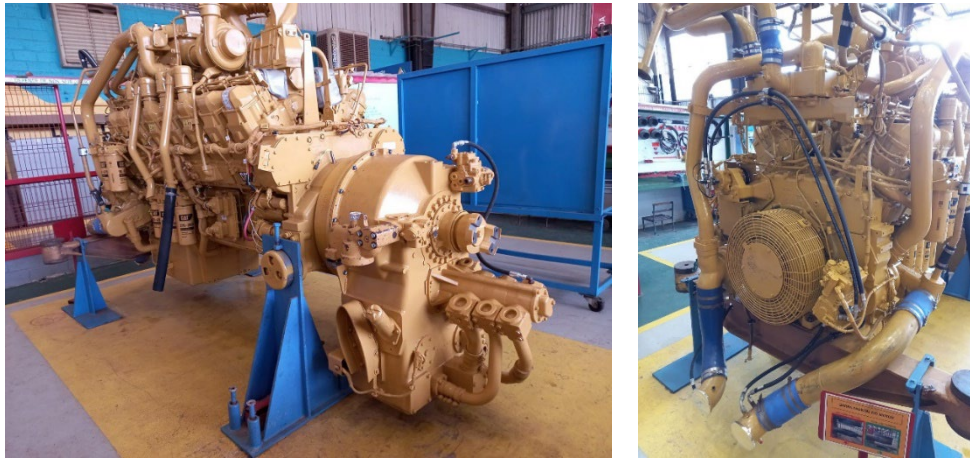


Figure 13, 14. Engine/ TC ready to Install on the truck

4.4.2 Radiator:

Radiator replacement time on an LMT is reduced by 9 hours (table / graphic 01) as the component will be ready to install it by performing:

- Full flushing;
- Core and seals replacement;
- Installation of fan hydraulic motor;
- Installation of fan and all guards;
- Installation of handrails and stairway;



Figure 15, 16. Radiator ready to Install on the truck

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4.4.3 Front Wheel, Brake and Suspension Assembly:

Front Wheel, Brake and Suspension Assembly replacement time on an LMT is reduced by 80 hours (table / graphic 01) as the front suspension will be ready to install it by performing:

- Brake and Bearings replacement;
- Front wheel and duo cone seals installation;
- Bearing pre-load procedure with more reliability (Due to spindle shaft is on vertical position);



Figure 17, 18. Front Wheel, Brake and Suspension ready to Install on the truck

5.0 Resources Required

Hand tools, Lift devices, Benches, Stands, Suitable reservoirs, Special tooling, Torquimeters, Parts needed, Spare components and Kits for repairs.

6.0 Supporting Attachments / References

Engine and Torque Converter - Separate and Connect (REN8334-04)
 Fan - Remove and Install (REN8334-04)
 Radiator - Disassemble (REN8334-04)
 Front Wheel Bearings - Install - Standard Wheel (KENR8579-14)

7.0 Related Best Practices

0107-2.3-1052 - Major Component Installation Kits Improve R&I Process

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